

Forecasting Voltage Fluctuations in Smart Grids Using CNN-Transformer Hybrid Models

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Abstract—This paper investigates short-horizon forecasting of distribution-grid RMS voltage using the SoCal28Bus dataset, focusing on Bus 1 and Bus 2 measurements sampled at 10-second resolution. A reproducible preprocessing pipeline was implemented to align multi-node time series and construct a 24-feature input matrix. A CNN-Transformer hybrid model was developed to capture local temporal patterns via convolution and longer-range dependencies via self-attention, using a 10-minute lookback window (60 steps). Forecasting performance was evaluated primarily using RMSE and MAE reported in volts on six RMS voltage targets (Bus 1 and Bus 2 phases L1–L3) to support operational interpretation. Like-for-like baseline comparisons were conducted against persistence and ridge regression under the same chronological train/validation/test split. Results show the CNN-Transformer achieved competitive accuracy and low inference latency suitable for near real-time monitoring, outperforming persistence but not consistently surpassing ridge regression on this low-variance RMS task. A focused sensitivity study (kernel size and attention heads) and a longer-horizon evaluation (6, 12 and 18 steps) further examined design choices and practical trade-offs. The findings emphasise the importance of strong baselines and volts-based reporting when assessing forecasting methods for stable distribution voltage signals.

Index Terms—Smart grid, voltage forecasting, CNN-Transformer hybrid, deep learning, time series prediction

I. INTRODUCTION

Electricity networks are shifting from centralised grids towards decentralised smart grids. The transition will be driven by renewable energy sources. Solar energy and wind energy, the increased deployment of distributed generation, and the digitalisation of power systems drive it [1]. Smart grids are energy systems with monitoring and control capabilities enabled by state-of-the-art information and communication technologies. They communicate in two directions between utilities and consumers to improve efficiency, reliability and flexibility [2]. Renewable sources are variable and intermittent, introducing voltage instability and operational uncertainty [3]. Voltage fluctuations affect power quality since they risk damaging equipment or disrupting supply. These challenges require predictive tools that anticipate such changes even before they occur [4], making voltage forecasting essential for the stability and reliability of modern smart-grid systems.

Voltage stability is defined as the ability of a power system to maintain acceptable voltage magnitudes at all buses in the

system during normal operating conditions and after being subjected to a disturbance. Each node in the system or bus is characterised by generators, loads and transmission lines [5]. The voltage in AC networks is represented by the phasor, which shows its value and phase angle. It is one of the two variables representing the power flowing between the nodes. The voltage is kept below a predefined maximum value, as otherwise the loss is induced [3]. With greater reliance on renewable generation and electric-vehicle loads, however, system voltage has become more nonlinear and harder to predict. An abrupt change in generation output or load demand can cause a jump in voltage on one part of the power grid to be felt across other sections [1]. If forecasts are accurate, the operator may be able to observe different voltage transients and deal with them pre-emptively to maintain safe and stable operation.

Conventional forecasting methods have focused on mathematical and statistical models based on the assumption of constant operating conditions and linear relationships between the system variables, such as autoregressive models or regression models; these methods are suitable for classical, centralised power systems [6]. As modern power systems tend to be highly dynamic, with distributed generation and non-linear demand patterns, statistical approaches often require important amounts of prior knowledge of the system parameters [5]. Furthermore, the difficulty of adaptively responding to changes in structure and environment limits the real-time implementation of statistical approaches [7].

This has led to the increasing use of artificial intelligence (AI) and machine-learning methods to learn from historical data [2]. AI-based methods have greater flexibility, scalability and adaptability to the different forecasting problems encountered in a modern smart grid.

Deep learning methods are widely used in voltage forecasting because they can learn useful patterns directly from large time-series datasets. Convolutional neural networks (CNNs) can capture short-term temporal patterns and relationships between signals, while recurrent networks such as LSTM can learn longer dependencies. However, both CNNs and LSTM can struggle when the context window becomes long. Transformer-based attention can help by modelling longer-range dependencies, which is important for voltage behaviour

that is influenced by both local and wider network conditions.

A more recent development for time-series prediction is the Transformer model, which uses a self-attention mechanism to capture all dependencies between steps without intrinsic ordering [7]. This makes it better at capturing long-term dependencies than RNN-based methods [5]. Transformer-based models have been applied to load forecasting and other power-system time-series prediction tasks with promising results. However, the models also have difficulty localising small variations amongst the voltage signals [2]. This is a challenge for CNNs, for which the features are currently based on local information from the grid data. A hybrid CNN and Transformer framework combines both local learning and global learning abilities [1]. This might be a stronger model for predicting voltage fluctuation in smart grids.

II. RELATED WORK

This section reviews relevant literature on short-term voltage forecasting in smart grids through hybrid deep learning techniques, identifying gaps that justify this work.

A smart grid is a modern power grid system that integrates renewable sources of energy, distributed storage systems, and control systems for efficient and sustainable operation [1]. In smart grids, two-way power flows can lead to voltage instability, due to the uncertainty of renewable generation and load behaviour [3]. Voltage dips, swells and fluctuations damage equipment, waste energy and cause power quality events, particularly in low voltage distribution networks. Therefore, short-term voltage prediction is an important research topic, especially for operational purposes and network control.

Early use of ML has shown voltage deviations can be predicted from past measurements without the need for knowledge of the network. Ataee et al. [16] used Random Forest regressors and classifiers to predict over- and under-voltages for a Swiss low-voltage distribution network with an accuracy of 85 per cent. Zufferey et al. [17] applied Quantile Neural Networks and k-nearest neighbours to improve the quality of state forecasting for distribution grids, reducing the voltage uncertainty and risk on the Basel distribution network. These efforts validated statistical learning as a viable option for managing voltage in real distribution grids.

Furthermore, the variability of renewable generation requires models to consider nonlinear spatio-temporal correlations [6]. With the advent of Phasor Measurement Units (PMUs) and advanced metering infrastructure, large amounts of data have been generated that often challenge existing analysis techniques. Since data-rich environments build rich models, emphasis should be on computationally efficient and interpretable models as part of a systematic review of forecasting models, as the operational trustworthiness and voltage stability of the grid is paramount under conditions of uncertainty.

Deep learning models are widely used in energy prediction tasks because they can learn hierarchical representations of highly complex data, such as Convolutional Neural Networks (CNN) for spatial dependencies and for short-term temporal

correlations [8]. Long Short-Term Memory (LSTM) networks, a type of Recurrent Neural Network (RNN) that can learn sequential dependencies between elements of a time series, have been proposed for load or voltage prediction [5]. This reflects the strength of these architectures across the energy domain.

By contrast, Gupta et al. [8] reported low percentage errors (e.g., MAPE below 2 per cent) for short-term forecasting tasks using CNN-LSTM combinations. However, percentage metrics such as MAPE can be sensitive to scaling choices and to small denominators; therefore, many power-system studies prioritise absolute error metrics (RMSE and MAE) reported in physical units where possible. In this paper, accuracy is evaluated primarily using RMSE and MAE in volts to support a direct operational interpretation.

However, CNN-RNN architectures have a number of drawbacks, including limited scalability and interpretability, and the recurring structure reduces parallelisation, making training expensive. Their intrinsic requirement for a fixed-size time window prevents them from capturing long-term dependency of the voltage patterns. Attention-based architectures, which allow the model to learn long-term temporal dependencies, instead of relying on recurrence, have been used in deep learning research, leading to the usage of Transformer-based architectures, which are more scalable and context aware for complex time series.

The introduction of the self-attention mechanism in Transformer networks transformed sequence modelling by allowing each time step's dependencies with each other to be learned in parallel [7], removing the sequential bottleneck of recurrent nets and allowing faster and more efficient learning from big data. Its success in natural language processing has led to use across power-system forecasting tasks (e.g., load and voltage prediction).

L'Heureux et al. [9] applied a Transformer-based approach to multi-regional load data achieving an average MAPE of 0.09, outperforming classical recurrent neural networks (RNN) and long short-term memory (LSTM) models. Wang et al. [7] showed the application of attention mechanisms in an RNN can improve the learning of temporal correlations for short-term load forecasting, improving robustness against missing data. Likewise, Li et al. [5] found that self-attention better captured long term dependencies between voltage and load data than recurrent models and led to more stable predictions, while Ahmed et al. [2] added context embeddings to further improve accuracy and interpretability.

Despite their ability to model temporal features, Transformers may not be the best architecture to model spatial features, such as the local features needed for specifying the voltages between a bus and its neighbours, and may be less sensitive to small voltage differences between buses. This has led to the development of hybrid CNN-Transformer models for time series forecasting to capture local patterns and temporal attention. Such hybridisation seeks to combine the best of both models, making both strong short-term predictions and generalisations over patterns that occur over longer time scales.

Hybrid architectures combine the localised feature extraction of CNNs with the long-range attention modelling of Transformers. CNN layers are used to extract spatial and short-term temporal features from the data. The attention mechanism of Transformer layers is responsible for modelling the long-term correlations [2]. For grid forecasting, Aksan et al. [10] show that adding CNNs before the attention modules is highly helpful as it reduces the models' sensitivity to noises and improves their learning stability.

For multi-scale load forecasting, Tian et al. [11] proposed a hybrid CNN-Transformer architecture and proved a 12 per cent RMSE improvement and 15 per cent training time reduction over standalone Transformers. Ullah et al. [12] proposed a BiGRU-CNN hybrid achieving under 1 per cent MAPE on large utility datasets, outpacing standalone models. These studies show the predictive power and efficiency of these hybrid designs for energy applications.

Sarker et al. [13] applied CNN, GRU and attention-based architecture in federated learning and explainable-AI settings and demonstrated that it achieved better MAE across different datasets. In addition, by deriving the feature importance scores through the SHAP method, they showed that explainable hybrid forecasting is indeed possible in distributed grids. However, hybrid approaches mostly focus on load forecasting rather than voltage forecasting. Because of the reactive component and higher frequencies in the voltage data, its prediction is harder. There are fewer studies on the influence of convolution kernel width, attention heads, and embedding dimension on voltage forecasting accuracy, which highlights the methodological gap this study seeks to address.

III. METHODOLOGY

This chapter discusses the methodology that was adopted to achieve the objectives of this research, namely predicting voltage fluctuations in the smart-grid environment through a CNN-Transformer hybrid model and comparing its performance with other models from previous literature. It explains the philosophical assumptions of this study, as well as the approach and methods applied in collecting and validating the data. The methodology is quantitative and computational, using a comparative experimental design. It applies supervised learning to time-stamped measurements from the SoCal28Bus smart-grid dataset. Although the dataset contains 28 buses, experiments in this dissertation use Bus 1 and Bus 2 as representative nodes; the same pipeline generalises to additional buses.

A. Dataset

The data for this research is drawn from the SoCal28Bus dataset. For reproducible experiments, the analysis uses two buses (Bus 1 and Bus 2) and focuses on phase RMS voltages as the forecasting targets, with frequency and lagged voltages used as explanatory features.

The dataset has also been made publicly available and has been used in studies of grid stability and predictive maintenance [?], [?].

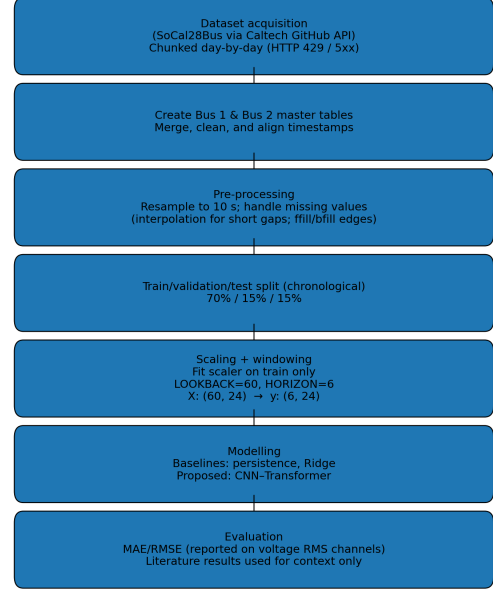


Fig. 1. Research flow for dataset preparation, modelling, and evaluation.

The details of each record include:

- Bus identifier (1-28).
- Timestamp.
- Per-phase voltage measurements (V_{L1} , V_{L2} , V_{L3} in volts).
- Per-phase frequency (Hz), RMS voltage, and phase angle.
- For each bus and each phase (L1-L3), the selected features are: voltage (V), frequency, RMS, and phase angle. This yields 12 features per bus and 24 features for Bus 1 + Bus 2.

The dataset spans across multiple days and contains a high volume of time-stamped samples per bus, forming a multi-variate time series. In the analysed period used for modelling (Bus 1 and Bus 2), RMS voltages remain close to nominal and vary within a narrow band (approximately 276–285 V), which is consistent with stable grid operation.

B. Data Pre-processing

Pre-processing is the process of converting the raw SoCal28Bus measurements into a format usable by time-series forecasting algorithms. Because deep-learning models are sensitive to scale, sequence length, and completeness, data were pre-processed using cleaning, normalisation, feature selection, and windowing, while preserving the temporal ordering of all data.

After merging the two buses on timestamp, missing values were handled using forward-fill and back-fill, followed by dropping any remaining rows with missing entries. This approach preserves temporal order and keeps the pipeline sim-

ple and reproducible. Non-numeric values or obvious sensor failures were removed during cleaning.

No aggressive outlier removal was applied because genuine grid events can appear as rapid voltage changes. Instead, basic sanity checks were used to remove clearly invalid entries (e.g., missing, infinite, or non-numeric values).

To standardise scaling, Min-Max normalisation is applied to all bus values between 0 and 1: The scaler is fitted on the training split only and then applied unchanged to the validation and test splits.

Feature choice was guided by the forecasting objective and the channels consistently available across Bus 1 and Bus 2. For each bus, the input matrix contains 12 time-synchronised features, yielding 24 features for the combined multi-node sequence. The prediction targets are the six three-phase RMS voltage channels (Bus 1 L1–L3 and Bus 2 L1–L3). The remaining channels in the 24-feature matrix are retained as explanatory inputs.

A sliding window is applied to convert the continuous time series into supervised samples. With a 10-second sampling interval, the lookback window is set to 60 time steps (10 minutes of history). The forecasting horizon is set to 6 time steps, corresponding to a 1-minute-ahead multi-step forecast. Each training sample therefore maps an input sequence X of shape (60, features) to an output sequence y of shape (6, targets), where targets are the three-phase RMS voltages at Bus 1 and Bus 2.

After resampling and alignment, the combined Bus 1 and Bus 2 series contained 43,992 aligned 10-second timestamps. After windowing with a 60-step lookback and 6-step horizon, 43,927 supervised samples were obtained. Samples were split chronologically into training (70%), validation (15%), and testing (15%). Scaling parameters were learned from the training split only and applied unchanged to validation and test splits.

C. CNN-Transformer Hybrid Model

The CNN-Transformer hybrid architecture solves the problem by using the feature-localising capacity of CNNs and the contextual learning capability of Transformers [?].

The three main components of the architecture: a convolutional feature extractor, a Transformer encoder, and a dense output layer.

Figure 2 illustrates the main components and data flow: a convolutional feature extractor, a Transformer encoder, and a dense output head that produces a multi-step forecast. Although the model predicts all 24 channels, performance is reported on the voltage RMS variables.

- **Convolutional Block:** Two 1-D convolutional layers with 64 and 128 filters with a kernel size of 3 extract short-term temporal features and phase relationship across time steps. The two layers use ReLU activation functions and Batch Normalisation. A Max-Pooling layer is used to compress the temporal dimension by a factor of 2.
- **Transformer Encoder Block:** Series of multi-head self-attention heads (8 heads) and feed-forward networks

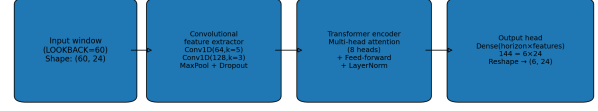


Fig. 2. CNN-Transformer architecture for multi-step multi-node forecasting.

(512 units) model the long-range dependencies in the sequence. The addition of positional encoding allows the model to take into account the location of the tokens. Residual connections along with layer normalisation aid gradient flow.

- **Fully Connected Layer (Dense):** After temporal pooling, dense layers map the learned representation to the forecast output. Dense layers of size 256 and 128 are used, followed by an output layer with size (horizon $\times$ features) = $6 \times 24 = 144$. The output is reshaped to (6, 24) to provide a multi-step forecast.

Each input window is a 3D tensor of shape (batch_size, sequence_length = 60, features = 24). The model predicts a multi-step sequence with horizon = 6, producing an output tensor of shape (batch_size, 6, 24). For reporting and discussion, results focus on the per-phase voltages at each bus (bus1_V_L1-3 and bus2_V_L1-3), since these are the primary variables of interest for voltage stability.

The model is trained for up to 100 epochs with early stopping (monitoring validation loss, patience = 15, restoring the best weights). The Adam optimiser is used with a learning rate of 0.001, and mean squared error (MSE) is used as the training loss. Training uses a mini-batch size of 32. Dropout is applied within the CNN and Transformer blocks to reduce overfitting. The best model checkpoint is selected based on the lowest validation loss.

To prevent overfitting, given the model's capacity, the following techniques were employed:

- Dropout of probability 0.2 to 0.3 after dense layers.
- L2 weight decay (1×10^{-4}).
- Use early stopping based on validation loss.
- Random temporal shifts of up to ± 2 samples are used for data augmentation.

D. Evaluation Metrics

Model evaluation is critical for understanding predictive reliability and confirming that the CNN-Transformer architecture is at least as good as existing models. To quantitatively evaluate forecasting accuracy, we adopt a set of benchmark statistical accuracy metrics commonly used in the power systems forecasting literature. These include Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE) and Coefficient of Determination

(R^2). Each of these metrics gives another perspective on model accuracy.

These metrics can be mathematically defined as:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (1)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (2)$$

$$MAPE = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (3)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (4)$$

where y_i , $\hat{y}_i$, and $\bar{y}$ are the actual, predicted, and mean of the observed voltages, respectively, and N is the number of samples.

IV. RESULTS AND ANALYSIS

A. Baseline Performance

Two simple baselines were implemented. The first was a persistence baseline that predicts each future voltage as the last observed value in the input window. The second baseline was a ridge regression model trained on flattened input windows to provide a stronger linear comparator while remaining computationally efficient.

As expected, ridge regression improved upon the persistence baseline by exploiting correlations across the full historical window. Table I reports overall MAE and RMSE across the six voltage targets and six-step horizon.

TABLE I
OVERALL TEST-SET PERFORMANCE FOR BASELINE MODELS

Model	MAE (V)	RMSE (V)	R^2
Persistence (last-value)	0.346	3.675	-0.028
Ridge regression	0.362	2.910	0.356
CNN-Transformer (proposed)	0.524	3.482	0.077

Table II provides a breakdown of MAE and RMSE per target.

TABLE II
BASELINE PERFORMANCE BY BUS AND PHASE (V)

Target (bus_phase)	Persistence			Ridge		
	MAE	RMSE	R^2	MAE	RMSE	R^2
Bus1_L1	0.3789	3.2569	0.3025	0.4442	2.8228	0.4760
Bus1_L2	0.3801	3.2583	0.2979	0.4598	2.8262	0.4718
Bus1_L3	0.3751	3.2555	0.2990	0.4466	2.8227	0.4730
Bus2_L1	0.3141	4.0634	-0.5505	0.2746	3.0044	0.1523
Bus2_L2	0.3140	4.0449	-0.5790	0.2749	2.9897	0.1373
Bus2_L3	0.3140	4.0429	-0.5692	0.2749	2.9873	0.1433

B. CNN-Transformer Performance

The proposed model achieved $RMSE = 3.482$ V and $MAE = 0.524$ V on the test set, with $R^2 = 0.077$. Relative to the baselines reported in Table I, this corresponds to a small improvement over persistence ($RMSE = 3.675$ V) but remains above ridge regression ($RMSE = 2.910$ V). Training completed in approximately 11.33 minutes on an NVIDIA T4 GPU, and inference averaged about 0.5 ms per test sample.

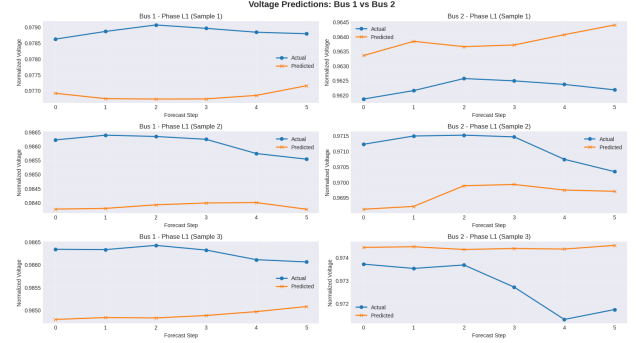


Fig. 3. multi-step predictions for Bus 1 and Bus 2 (RMS phases).

Per-target performance is summarised in Table III and indicates consistent accuracy across phases, with slightly larger errors on Bus 1 than Bus 2. On average, Bus 1 achieved mean $RMSE = 3.816$ V, while Bus 2 achieved mean $RMSE = 3.113$ V.

TABLE III
PER-BUS AND PER-PHASE RMS PERFORMANCE FOR THE
CNN-TRANSFORMER MODEL (V)

Bus	Phase	RMSE (V)	MAE (V)	R^2
Bus 1	L1	3.810	0.692	0.045
Bus 1	L2	3.826	0.745	0.032
Bus 1	L3	3.811	0.695	0.039
Bus 2	L1	3.129	0.369	0.081
Bus 2	L2	3.105	0.315	0.069
Bus 2	L3	3.106	0.329	0.074

C. Sensitivity Analysis

To address design parameter influence, a small sensitivity study was conducted over convolution kernel size and the number of attention heads. The configuration with kernel size 3 and 2 attention heads achieved the lowest test RMSE (3.433 V) in this sweep. Increasing the number of heads to 4 tended to degrade performance, and the larger kernel (5) did not provide a consistent benefit. This suggests that, for stable RMS voltage series, a modest receptive field and limited attention capacity are sufficient to capture the dominant short-horizon dynamics, while additional capacity may introduce unnecessary variance.

D. Longer-Horizon Evaluation

To evaluate practical applicability, the forecasting horizon was increased from 6 steps (1 minute) to 12 and 18 steps while keeping the same lookback window and evaluation protocol. Across the tested settings, test RMSE decreased from 3.976

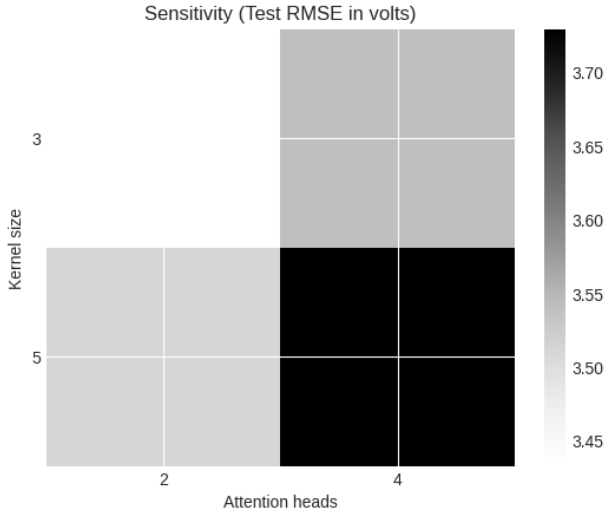


Fig. 4. Sensitivity heatmap showing test RMSE (V) across kernel size and attention heads.

V at 6 steps to 3.822 V at 12 steps and 3.659 V at 18 steps. This should be interpreted with caution. RMS voltage exhibits low variance and the horizon models may converge toward conservative forecasts that minimise average absolute error.

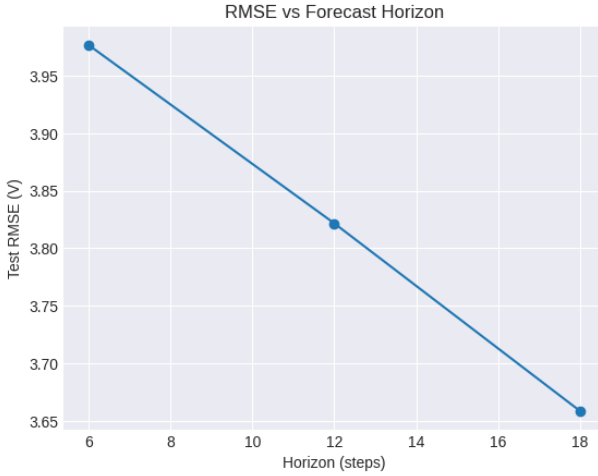


Fig. 5. Test RMSE (V) as a function of forecast horizon (steps).

V. CONCLUSION

This paper investigated short-horizon forecasting of distribution-grid RMS voltage using the SoCal28Bus dataset, focusing on Bus 1 and Bus 2 measurements. A consistent pre-processing pipeline was implemented to align multi-node time series at 10-second resolution and construct a 24-feature input matrix. A CNN-Transformer hybrid model was implemented to learn short-term patterns using convolutional layers and to integrate information across the full lookback window using self-attention.

The empirical results indicate that the proposed CNN-Transformer can produce accurate short-horizon RMS voltage

forecasts, but that simple linear baselines remain highly competitive for this dataset. On the voltage-only evaluation (six RMS channels), the CNN-Transformer achieved an overall test RMSE of 3.482 V and MAE of 0.524 V. This improved on the persistence baseline (RMSE 3.675 V), but did not outperform ridge regression (RMSE 2.910 V).

The sensitivity study showed that the best-performing configuration used kernel size 3 with 2 attention heads (test RMSE 3.433 V). Increasing the number of heads to 4 generally increased error, and the larger kernel (5) did not provide a consistent advantage.

From an efficiency perspective, the main model training run completed in approximately 11.33 minutes on an NVIDIA T4 GPU, and inference latency was around 0.5 ms per test sample. This indicates that the approach remains practical for frequent short-horizon updates in operational monitoring pipelines.

The work supports a careful, evidence-led approach to model selection, where complexity is matched to the dynamics of the operational signal and validated against strong, transparent baselines.

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